Sampling-based zero-order derivatives for learning reachable sets in viscous Hamilton-Jacobi PDEs

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***Abstract*— Computing the safe sets for dynamic systems in reachability settings, parameterized by the viscous Hamilton-Jacobi equation in general, involves problem space discretization on a grid to achieve a numerically consistent solution that is robust with respect to the original non-viscous solution. This presents a two-fold problem: discretization implies exponential memory requirements; and, computational costs increase cubically as problem dimensions increase. While parallel computing can reduce the impact of the latter problem, it does not address the exponential memory requirement. Leveraging a recent connection between HJ value functions and procimity operators, we introduce a moving-average Gaussian sampling scheme that computes the proximity operator for the smooth spatial gradients of the HJ value in a reachability context. This circumvents the upwinding schemes that are in popular use today and provides a computationally-efficient means of resolving safe sets in a time-efficient manner without dealing with the exponential memory requirements that grid discretization methods provide. Our hypothesis and mathematical machinery is backed up by numerical examples that demonstrate the feasibility of our approach on practical safety-verification problems.**

I. Introduction

The results of this paper take a further step towards tractable and scalable solutions to viscous Hamilton-Jacobi (HJ) [1] problems for ascertaining the verification (freedom from harm) [2] of *high-dimensional* dynamical systems. We work within a viscous HJ reachability analyses context [3]–[6] and establish our results with approximate globally optimal solutions to the computational problems. Our mainstay is the first-order nonlinear scalar Hamilton-Jacobi (HJ) partial differential equations (PDE) of the form

$$\begin{aligned} &\boldsymbol{v}_t + \boldsymbol{H}(t; \boldsymbol{x}, \nabla_x \boldsymbol{v}) = 0 \text{ in } \mathbb{R}^n \times (0, T] \\ &\qquad \boldsymbol{v}(\boldsymbol{x}, 0) = \boldsymbol{g}(\boldsymbol{x}) \text{ on } \mathbb{R}^n \times t = 0 \end{aligned} \tag{HJ-IVP}$$

where $\boldsymbol{v}_t$ denotes the partial derivative of the vector field $\boldsymbol{v}(\boldsymbol{x}, t)$ with respect to t; the Hamiltonian $\boldsymbol{H}(\boldsymbol{x}, t)$ is continuously defined on $\mathbb{R}^n$; $\boldsymbol{g}(\boldsymbol{x})$ is bounded and uniformly continuous (BUC) in $\mathbb{R}^n$; and $\nabla_x \boldsymbol{v}$ is the spatial gradient of $\boldsymbol{v}(\boldsymbol{x}, t)$. While a unique C^2 solution $\boldsymbol{v}(\boldsymbol{x}, t)$ exists uniformly on $0 \le t < T$ to (HJ-IVP) for which $\lim_{t \to T} \boldsymbol{v}(\boldsymbol{x}, t)$ (when $\boldsymbol{H}$ and $\boldsymbol{g}$ are smooth and $\boldsymbol{v}_0(\boldsymbol{x}, t)$ are compactly supported), $\boldsymbol{H}(\boldsymbol{x}, t)$ lacks continuous derivatives in this limit so that $\nabla_x \boldsymbol{v}$ *vanishes* at $t = T$.

In this article, we only care for *approximate solutions*. One may find Lipschitz continuous functions $\boldsymbol{v}(\boldsymbol{x}, t)$ on $\mathbb{R}^n \times (0, T]$ which are smooth *almost everywhere* (a.e.) and satisfy (HJ-IVP) in a "generalized sense". Suppose that we fix a positive $\delta > 0$, Crandall and Lions [7] showed that if $\delta > 0$, then $\boldsymbol{v}^\delta$ is the *vanishing viscosity* solution of

$$\begin{aligned} &\boldsymbol{v}^\delta_t + \boldsymbol{H}(t; \boldsymbol{x}, \nabla_x \boldsymbol{v}^\delta) = \frac{\delta}{2} \Delta \boldsymbol{v}^\delta \text{ in } \mathbb{R}^n \times (0, T] \\ &\boldsymbol{v}^\delta(\boldsymbol{x}, 0) = \boldsymbol{g}(\boldsymbol{x}) \text{ on } \mathbb{R}^n \times t = 0 \end{aligned} \tag{HJ-Viscous}$$

with Laplacian $\Delta \boldsymbol{v}^\delta$ and a constant $k > 0$. Uniform convergence of $\boldsymbol{v}^\delta \to \boldsymbol{v}$ exists in the limit $\delta \to 0^+$, i.e.

$$\sup_{t \in (0, T)} \sup_{x \in \mathbb{R}^n} \mid \boldsymbol{v}(\boldsymbol{x}, t) - \boldsymbol{v}^\delta(\boldsymbol{x}, t) \mid \le k\sqrt{\delta}. \tag{1}$$

Notice that equation (HJ-Viscous) models conservation laws in a single spatial dimension [8]. For inf-sup or sup-inf optimal control problems [3], the Hamiltonian is related to the *backward* reachable set [4] of a dynamical system. Reachable systems are those where one can compute all states that can be reached from an initial condition in *a finite number of steps*.

Popular numerically consistent optimal solutions [7], [9] to (HJ-Viscous) utilize problem space discretization on a grid [4], [10]. However, these require exponential memory storage – making their practical applications limited for large problems. Our recent efforts [5], [6] leveraged single-instruction, multiple threads (SIMT) computing schemes on modern GPUs to accelerate computational times on a host of practical problems; we demonstrated improved scalability with up to $76 - 92\%$ improvement in computational times on higher-dimensional problems. However, it did not resolve the memory requirements that a grid introduces in arbitrarily large dimensions without resorting to grids bifurcation. We thus tack another thread in this article, jettisoning grid discretization methods for proximal operators that resolve the global extrema of the Moreau envelopes of HJ equations. These provide global solutions in nonconvex optimization settings [11]–[14] towards improving the computational efficiency of approximate solutions to viscous HJ PDEs for practical problems. It remains to demonstrate the applications of these schemes to reachable sets within the context of HJ theory.

The Cole-Hopf transformation [1], [13] employed in PDE analysis relates the heat equation solution to that of the viscous HJ equation. Recently, Osher et. al [8] applied the Cole-Hopf transformation to $\boldsymbol{v}^\delta$ and related the resulting transformed heat conservation equation to an importance sampling scheme for approximating the solution $\boldsymbol{v}^\delta$ of (HJ-Viscous). We build on this, introducing a sample-efficient normal-inverse-Wishart moving-average sampling

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scheme for evaluating the spatial gradients of the value function in order to resolve *robust controlled backward reachable sets and tubes* [15] in the terminal Hamilton-Jacobi-Isaacs equation [4], [5], [16].

The rest of this paper is structured as follows: in §II, we provide the theoretical machinery for our problem. In §III, we describe our methods. Section IV presents our numerical algorithms on sample problems and we conclude the paper in §V.

II. Background

We first introduce the notations that are commonly througout this article. Reachable sets within the context of two person games [16] and their accompanying "viscous" terminal HJI PDE [17] are then introduced. This is followed by the variational HJ-Isaacs PDE needed to resolve the sublevel set that characterizes a protoyypical reachable set.

A. Notations and Terminologies

Conventions: Bold font upper-case and lower-case Roman letters are matrices and vectors, respectively; calligraphic letters are sets. Exception: the subdifferential *set* of a BUC function, $\boldsymbol{g}(\boldsymbol{x})$, is denoted ϑ. Time variables e.g. t, τ, T will always be positive real numbers. The n-dimensional Euclidean space is $\mathbb{R}^n$ and the set of natural numbers is denoted $\mathbb{N}$. The natural log of T is denoted $\log T$. The dot product is denoted $\langle \cdot, \cdot \rangle$. The partial time derivative of $\boldsymbol{v}$ is written $\partial \boldsymbol{v}_t$ while its partial spatial derivative is denoted $\partial \boldsymbol{v}_{\boldsymbol{x}}$. For a function $f(x(t), y(t))$ with every argument parameterized by t, we simply write $f(t; x, y)$. We represent the real-valued Hilbert space with $\boldsymbol{\mathcal{H}}$; denote $2^{\boldsymbol{\mathcal{H}}}$ by its power set, i.e. the family of all subsets of $\boldsymbol{\mathcal{H}}$. Associated with $\boldsymbol{\mathcal{H}}$ is the maximally monotone set-valued operator, $\boldsymbol{Q} : \boldsymbol{\mathcal{H}} \to 2^{\boldsymbol{\mathcal{H}}}$. The zeroes of $\boldsymbol{Q}$ are denoted zer $\boldsymbol{Q} = \{\boldsymbol{x} \in \boldsymbol{\mathcal{H}} \mid 0 \in \boldsymbol{Q}\boldsymbol{x}\}$. The set of global minimizers of $\boldsymbol{g}(\boldsymbol{x})$ is denoted $\texttt{Argmin}\ \boldsymbol{g}(\boldsymbol{x})$. It is written as $\operatorname{argmin} \boldsymbol{g}(x)$ when the set is a singleton. The set of fixed point of operator $S : \boldsymbol{\mathcal{H}} \to \boldsymbol{\mathcal{H}}$ is $\text{Fix } S = \{\boldsymbol{x} \in H \mid S\boldsymbol{x} = \boldsymbol{x}\}$.

The unique solution $\boldsymbol{x} \in \mathbb{R}^n$ of the dynamical system $\dot{\boldsymbol{x}}(\tau) = f(t; \boldsymbol{x}, \boldsymbol{u}, \boldsymbol{w})$ is influenced by a pursuing player $\boldsymbol{P}$ (with control $\boldsymbol{w} \subseteq \mathbb{R}^p$) and its evading pair $\boldsymbol{E}$ (with control $\boldsymbol{u} \subseteq \mathbb{R}^m$). System trajectories $\xi_{\boldsymbol{x},\tau}^{\boldsymbol{u},\boldsymbol{w}}$ emanate from $\dot{\boldsymbol{x}}(\tau)$ by minimizing a scalar-valued objective

$$\mathcal{J}(\tau; \boldsymbol{x}, \boldsymbol{u}, \boldsymbol{w}) = \min_{\tau \in [t,T]} \boldsymbol{g}(\xi_{\boldsymbol{x},t}^{\boldsymbol{u},\boldsymbol{w}}(\tau)) \qquad \text{(Target set cost)}$$

within a two-person differential game for a fixed $T > \tau > t$. The controllers belong in compact sets which are measurable functions

$$\bar{\mathcal{U}} \equiv \boldsymbol{u} : [t, T] \to \mathcal{U}, \qquad \bar{\mathcal{W}} \equiv \boldsymbol{w} : [t, T] \to \mathcal{W}. \qquad (2)$$

B. Lower value of the differential game

For a (possibly infinite) *terminal time* T, a constant k, all $\hat{\boldsymbol{x}}$, $\boldsymbol{x} \in \mathbb{R}^n$, $\boldsymbol{u} \in \mathcal{U}$ and $\boldsymbol{w} \in \mathcal{W}$, let the system possess a BUC payoff functional $\boldsymbol{g}(\boldsymbol{x}(T))$ that satisfies $\boldsymbol{g} : \mathbb{R}^n \to \mathbb{R}$. The differential game's *lower value* [18]

$$\boldsymbol{v}^-(\boldsymbol{x}, t) = \inf_{\beta \in \mathcal{B}(t)} \sup_{\boldsymbol{u} \in \mathcal{U}(t)} \mathcal{J}(t; \boldsymbol{x}, \boldsymbol{u}, \boldsymbol{w}) \qquad (3a)$$

$$\triangleq \inf_{\beta \in \mathcal{B}(t)} \sup_{\boldsymbol{u} \in \mathcal{U}(t)} \min_{\tau \in [t,T]} \boldsymbol{g}(\xi_{\boldsymbol{x},t}^{\boldsymbol{u},\boldsymbol{w}}(\tau)). \qquad (3b)$$

is the viscosity solution to (HJ-IVP) [18, Lemma 2.1] with Hamiltonian,

$$\boldsymbol{H}^-(t, \boldsymbol{x}, \boldsymbol{u}, \boldsymbol{w}, p) = \max_{\boldsymbol{u} \in \mathcal{U}} \min_{\boldsymbol{w} \in \mathcal{W}} \langle f(t, \boldsymbol{x}, \boldsymbol{u}, \boldsymbol{w}), p \rangle. \qquad (4)$$

Notice in (3a) that the pursuer possesses a non-anticipative strategy $\mathcal{B}(t)$ that is aware of all (maximizing) controls of the evading player up until time t. In what follows, we drop the negative superscript to refer to the value of the game in the context of reachability analysis.

C. Robustly Controlled Backward Reachable Set and Tube

Suppose that the goal of $\boldsymbol{E}$ is to reach a user-specified target region $\mathcal{L}_0(\tau)$ within T time steps of playing the game; while the $\boldsymbol{P}$ simultaneously seeks to prevent this from happening. Since we are in a two-player best-response scenario, the invariant set $\mathcal{L}_0$ obtained at T

$$\mathcal{L}_0(T) = \{\boldsymbol{x} \in \mathbb{R}^n \mid \boldsymbol{g}(0; \boldsymbol{x}) \leq 0\} \qquad \text{(Target-Set)}$$

is "*robustly controlled*" for the "*distance-to-target-set*" cost $\boldsymbol{g}(0; \boldsymbol{x})$, with constraints $|\boldsymbol{g}(0; \boldsymbol{x})| \leq k,\ |\boldsymbol{g}(0; \boldsymbol{x}) - \boldsymbol{g}(t; \hat{\boldsymbol{x}})\ | \leq k|\boldsymbol{x} - \hat{\boldsymbol{x}}\ |$ where $k > 0$ is a constant all $-T \leq t \leq 0$, $\{\hat{\boldsymbol{x}}, \boldsymbol{x} \in \mathbb{R}^n\}$[1]. The distance to $\mathcal{L}_0(\tau)$ is typically found by optimizing $\boldsymbol{g}(\boldsymbol{x}, t)$ as in (Target set cost). The HJI PDE (3a) and Hamiltonian (4) for this set admit the form

$$\begin{aligned} &\boldsymbol{v}_t(t; \boldsymbol{x}) + \min\{0, \boldsymbol{H}\,(x, \partial \boldsymbol{v}_{\boldsymbol{x}}(t; \boldsymbol{x}))\} = 0, \\ &\boldsymbol{v}(0; \boldsymbol{x}) = \boldsymbol{g}(0; \boldsymbol{x}). \end{aligned} \qquad \text{(HJI-RCBRT)}$$

For the safety problem setup in (3a), the corresponding *robustly controlled backward reachable **tube*** (RCBRT) on $(0, T]$ is the closure of the open set [15]

$$\begin{aligned} \mathcal{L}([\tau, 0], \mathcal{L}_0) = \{&\boldsymbol{x} \in \mathbb{R}^n \mid \exists\, \beta \in \bar{\mathcal{W}}(t)\, \forall\, \boldsymbol{u} \in \mathcal{U}(t), \exists \\ &\tau \in [-T, 0], \boldsymbol{\xi}(\bar{t}) \in \mathcal{L}_0\}. \end{aligned} \qquad \text{(Target-Tube)}$$

Read: The set of states from which the strategies of $\boldsymbol{P}$ and for all controls of $\boldsymbol{E}$ imply that we *reach and remain in the target set* in the interval $[-T, 0]$. Following Lemma 2 of [4], the states in the reachable set admit the following properties w.r.t the value function $\boldsymbol{v}^-$:

$$\boldsymbol{x}(t) \in \mathcal{L}(\cdot) \implies \boldsymbol{v}(\boldsymbol{x}, t) \leq 0, \qquad (5a)$$

$$\boldsymbol{v}(\boldsymbol{x}, t) \leq 0 \implies \boldsymbol{x}(t) \in \mathcal{L}(\cdot). \qquad \text{(Zero Levelset)}$$

In backward reach avoid tubes, which is typically the case in most practical sense, where the agent must avoid the unsafe region at all times, we can write (HJI-RCBRT) as the HJI

[1]Time is reversed in BRT computational scenarios.

variational inequality of the *robustly controlled backward reach-avoid tube* (RCBRAT) as follows,

$$\min\{\boldsymbol{v}_t(t;\boldsymbol{x})+\boldsymbol{H}\left(t;x,\partial\boldsymbol{v}_{\boldsymbol{x}}\right),\boldsymbol{g}(\boldsymbol{x},t)-\boldsymbol{v}(t;\boldsymbol{x})\}\leq 0,$$
$$\boldsymbol{v}(0;\boldsymbol{x})=\boldsymbol{g}(0;\boldsymbol{x}). \tag{HJI-RCBRAT}$$

Challenges in optimizing the RCBRT/RCBRAT: Computing the RCBRT presents a host of challenges for optimization because (a) the dynamics $f(\cdot)$ is usually nonlinear with coupled states in most problems of interest; (b) although $\mathcal{L}([\tau,0],\mathcal{L}_0)$ is mostly smooth, it possesses discontinuous "shocks" at solution crossover points (see [15]); this makes global gradient-based optimization impossible for resolving $\boldsymbol{v}$; (c) more than one trajectory can emanate from the RCBRT's boundary that end up at the same point on the target set – this makes sampling-based optimization difficult; and (d) the RCBRT is nonconvex.

To mitigate these challenges above, we introduce a set of tools that allows us to circumvent these challenges in an optimization context.

D. Cole-Hopf Transformation and the Heat Equation

For the BUC function $\boldsymbol{g}(t;\boldsymbol{x}):\mathbb{R}^n\to\mathbb{R}$ from (HJ-IVP), consider the transformation $\boldsymbol{\varphi}^\delta(\boldsymbol{x})=\exp\left(-\boldsymbol{v}^\delta/\delta\right)$ originally attributed to Cole and Hopf [1, §4.4.1]. Applying the Cole-Hopf transformation to (HJ-Viscous), we can write

$$\partial\boldsymbol{\varphi}_t^\delta(\boldsymbol{x},t)-\frac{\delta}{2}\boldsymbol{\varphi}^\delta(\boldsymbol{x},t)=0 \text{ in } \mathbb{R}^n\times(0,T]$$
$$\boldsymbol{\varphi}^\delta(\boldsymbol{x},t)=\exp\left(-\boldsymbol{g}/\delta\right) \text{ on } \mathbb{R}^n\times\{t=0\}^2. \tag{Heat-Kernel}$$

The fundamental solution, $\boldsymbol{\Phi}_{\delta t}$ to (Heat-Kernel) at time t is [1]

$$\boldsymbol{\Phi}_{\delta t}(\boldsymbol{x})=\begin{cases}(2\pi\delta t)^{n/2} & \exp\left(-\dfrac{|\boldsymbol{x}|^2}{2\delta t}\right) \text{ in } \mathbb{R}^n\times(0,T],\\ 0 & \text{otherwise .}\end{cases}$$

The heat equation (Heat-Kernel) admits the following explicit formulae (as a convolution, an integral expression, and a Gaussian expectation; see derivation in [11, Appendix B]),

$$\boldsymbol{\varphi}^\delta(t;\boldsymbol{x})=\left(\boldsymbol{\Phi}_{\delta t}\star\exp(-\boldsymbol{g}/\delta)\right)(\boldsymbol{x}), \tag{6a}$$
$$\equiv\int_{\mathbb{R}^n}\left(\boldsymbol{\Phi}_{\delta t}(\boldsymbol{x}-\boldsymbol{y})\exp(-\boldsymbol{g}(\boldsymbol{y})/\delta)\right)d\boldsymbol{y}, \tag{6b}$$
$$=\mathbb{E}_{y\sim\mathcal{N}(\boldsymbol{x},\delta t)}\left[\exp(-\delta^{-1}\boldsymbol{g}(\boldsymbol{y}))\right]. \tag{6c}$$

The equation (Heat-Kernel) transforms solutions to the heat equation [1], [12] to those of the viscous HJ PDE. It has enjoyed use in studying the energy landscape of neural networks [13], [19] and viscous HJ loss functions [11]. Inspecting (Heat-Kernel), we may write

$$\boldsymbol{g}^\delta(\boldsymbol{x}):=\boldsymbol{v}^\delta(t;\boldsymbol{x}) \tag{7a}$$
$$=-\delta\log\left(\boldsymbol{\varphi}^\delta(t;\boldsymbol{x})\right)(\boldsymbol{x}) \text{ in } \mathbb{R}^n\times(0,T], \tag{7b}$$

from which we can write

$$\partial\boldsymbol{v}_{\boldsymbol{x}}^\delta(t;\boldsymbol{x})=-\delta\cdot\frac{\partial\boldsymbol{\varphi}_{\boldsymbol{x}}^\delta(t;\boldsymbol{x})}{\boldsymbol{\varphi}^\delta(t;\boldsymbol{x})}. \tag{8}$$

Chaudhari et al. [13] called $\boldsymbol{g}^\delta(\boldsymbol{x})$ the local entropy and postulated a stochastic differential equation (SDE) scheme that leverages $\boldsymbol{g}^\delta(\boldsymbol{x})$ in SGD contexts in resolving the viscous HJ PDE. Recent publications treat HJ PDE optimization [12] in light of this local entropy function by combining Moreau envelopes [20] and proximal operator [21] algorithmics to establish zero-order optimization schemes for the viscous HJ loss function (with guaranteed convergence to global optima).

III. Methods

We now cast the computation of the optimal "controllable"[3] reachable set into a form that leverages the Moreau envelope (Moreau-Env) in a successive approximation scheme so that the reachable set can be obtained via the subdifferential operator of the problem.

A. Operators and Envelopes of HJ Equations

Monotone operator theory has had a huge impact on partial differential equations and nonlinear systems [22], [23], reducing nonlinear analyses problems to the form

$$\text{determine } \boldsymbol{x}\in \text{ zer } \boldsymbol{Q}=\{\boldsymbol{x}\in\boldsymbol{\mathcal{H}}\,|\,0\in\boldsymbol{Q}\boldsymbol{x}\}. \tag{9}$$

The subdifferential operator (see [21, §16]), $\partial\boldsymbol{g}:\boldsymbol{\mathcal{H}}\to 2^{\boldsymbol{\mathcal{H}}}$, of $\boldsymbol{g}(\boldsymbol{x})$ at $\bar{\boldsymbol{x}}$ is the set of all ϑ that satisfies

$$\{\vartheta\in\boldsymbol{\mathcal{H}}\mid(\forall\boldsymbol{x}\in\boldsymbol{\mathcal{H}})\langle\boldsymbol{x}-\bar{\boldsymbol{x}},\vartheta\rangle+\boldsymbol{g}(\bar{\boldsymbol{x}})\leq\boldsymbol{g}(\boldsymbol{x})\}. \tag{10}$$

That is, $\boldsymbol{g}$ is subdifferentiable at $\boldsymbol{x}$ if $\partial\boldsymbol{g}(\boldsymbol{x})\neq\emptyset$; hence, elements of $\partial\boldsymbol{g}(\boldsymbol{x})$ are the subgradients of $\boldsymbol{g}$ at $\boldsymbol{x}$.

An optimization implication of (10) is Fermat's rule, which states that for $\boldsymbol{g}:\boldsymbol{\mathcal{H}}\to(0,T)$ [24],

$$\text{argmin } \boldsymbol{g}=\text{zer}\,\partial\boldsymbol{g}. \tag{11}$$

The proximity operator of $\boldsymbol{g}(\boldsymbol{x})$ at a time, $t\in(0,T]$ is the minimizers set [25]

$$\text{Prox}_{t\boldsymbol{g}}(\boldsymbol{x})\triangleq\arg\min_{\boldsymbol{z}\in\boldsymbol{\mathcal{H}}}\left(\boldsymbol{g}(\boldsymbol{z})+\frac{1}{2t}\|\boldsymbol{z}-\boldsymbol{x}\|^2\right) \tag{Prox-Op}$$

that define the Moreau envelope $\boldsymbol{v}(\cdot,t)$, defined as [20]

$$\boldsymbol{v}(\boldsymbol{x},t)=\min_{\boldsymbol{z}\in\boldsymbol{\mathcal{H}}}\left(\boldsymbol{g}(\boldsymbol{z})+\frac{1}{2t}\|\boldsymbol{z}-\boldsymbol{x}\|^2\right). \tag{Moreau-Env}$$

The envelope $\boldsymbol{v}$ expands troughs of $\boldsymbol{g}(\boldsymbol{x})$ and its local minimizers concur with those of $\boldsymbol{g}$ [13].

Lemma 1: Suppose that $\boldsymbol{g}(\boldsymbol{x})$ is BUC, its subdifferential exists, the set $\mathcal{S}_\alpha\triangleq\{z\in\mathbb{R}^n:\boldsymbol{g}(\boldsymbol{z})\leq\inf\boldsymbol{g}+\alpha\}$ is compact, zer $\partial\boldsymbol{g}$ exists and $\boldsymbol{x}\in\mathcal{S}_\alpha$. An important result established in [11, Lemma A.3] relates the spatial gradients of the solution to (HJ-IVP) to the proximity operator of $\boldsymbol{g}(\boldsymbol{x})$ at time t as

$$\text{Prox}_{t\boldsymbol{g}}(\boldsymbol{x})=\boldsymbol{x}-t\partial\boldsymbol{v}_{\boldsymbol{x}}(t;\boldsymbol{x}):\approx\boldsymbol{x}-t\cdot\partial\boldsymbol{v}_x^\delta(t;\boldsymbol{x}). \tag{12}$$

[3]We use this term because the agents in the differential game system have control inputs

Our goal now is to leverage the Proximity operator in the minimization (Target set cost) while playing the zero-sum two-player differential game (HJI-RCBRAT) to optimize for the RCBRT (Target-Tube) and the zero levelset (5). In this light, we can replace $\boldsymbol{v}(\boldsymbol{x}, t)$ (HJ-IVP) by its viscous and smooth proxy (HJ-Viscous), evaluate the minimizer (Target set cost) using the proximal operator result (1) and solve for the `inf-sup` optimal control problem in (3a).

B. The Kernel of the HJ PDE and its proximity operator

Consider the kernel $\boldsymbol{\varphi}^\delta(\boldsymbol{x}, t)$ in (Heat-Kernel) which corresponds to the solution of (HJ-Viscous). Resolving the convolution (6) is intractable for large n and time step t. However, it is same as the probability density of a Gaussian distribution with mean $\boldsymbol{x}$ and variance δt (as derived in [11, Appendix B]), i.e.

$$\partial \boldsymbol{\varphi}^\delta_{\boldsymbol{x}}(t; \boldsymbol{x}) = -\frac{1}{\delta t} \cdot \mathbb{E}_{y \sim \mathcal{N}(\boldsymbol{x}, \delta t)} \left[(\boldsymbol{x} - \boldsymbol{y}) \exp(\boldsymbol{g}(\boldsymbol{y})/\delta) \right]. \tag{13}$$

Plugging (6c) and (13) into (8), we find that

$$t \cdot \partial \boldsymbol{v}^\delta_{\boldsymbol{x}}(t; \boldsymbol{x}) = \boldsymbol{x} - \frac{\mathbb{E}_{y \sim \mathcal{N}(\boldsymbol{x}, \delta t)} \left[\boldsymbol{y} \exp(-\boldsymbol{g}(\boldsymbol{y})/\delta) \right]}{\mathbb{E}_{y \sim \mathcal{N}(\boldsymbol{x}, \delta t)} \left[\exp(-\delta^{-1} \boldsymbol{g}(\boldsymbol{y})) \right]} \tag{14}$$

provides the time-scaled spatial gradients of $\boldsymbol{v}(t; \boldsymbol{x})$ in (17) without the need for unstableschemes employed in upwinding methods within grid discretization numerical reachable sets' computational algorithms [5], [6], [26].

Substituting the r.h.s of (14) into (1), we find that

Proximal Operator for $\nabla_{\boldsymbol{x}} \boldsymbol{v}^\delta(t; \boldsymbol{x})$

$$\text{Prox}_{t\boldsymbol{g}}(\boldsymbol{x}) = \frac{\mathbb{E}_{y \sim \mathcal{N}(\boldsymbol{x}, \delta t)} \left[\boldsymbol{y} \cdot \exp(-\boldsymbol{g}(\boldsymbol{y})/\delta) \right]}{\mathbb{E}_{y \sim \mathcal{N}(\boldsymbol{x}, \delta t)} \left[\exp(-\delta^{-1} \boldsymbol{g}(\boldsymbol{y})) \right]} \tag{15}$$

C. Hamilton-Jacobi–Proximity Operator Relation

Now consider the HJI PDE (HJI-RCBRT) and the HJI-PDE for the RCBRAT (HJI-RCBRAT) in light of (15). We can write the domain of these PDEs in light of the proximal expression (15) as

$$\begin{aligned}
&\partial \boldsymbol{v}_t(t; \boldsymbol{x}) + \min\{0, \boldsymbol{H}\left(x, \partial \boldsymbol{v}_{\boldsymbol{x}}(t; \boldsymbol{x})\right)\} = 0, \\
&\approx \partial \boldsymbol{v}^\delta_t(t; \boldsymbol{x}) + \min\{0, \boldsymbol{H}^\delta\left(t; x, \partial \boldsymbol{v}^\delta_{\boldsymbol{x}}\right)\} = 0, \\
&:= \partial \boldsymbol{v}^\delta_t(t; \boldsymbol{x}) + \min\left\{0, \max_{\boldsymbol{u} \in \mathcal{U}} \min_{\boldsymbol{w} \in \mathcal{W}} \left\langle f(t; \boldsymbol{x}, \boldsymbol{u}, \boldsymbol{w}), \right.\right. \\
&\qquad \left.\left. \frac{1}{t}(\boldsymbol{x} - \text{Prox}_{t\boldsymbol{g}}(\boldsymbol{x})) \right\rangle \right\} = 0,
\end{aligned} \tag{16}$$

for (HJI-RCBRT). Similarly, the right-hand-side of the variational inequality for (HJI-RCBRAT) can be written as

$$\begin{aligned}
&\min\{\partial \boldsymbol{v}_t(t; \boldsymbol{x}) + \boldsymbol{H}\left(t; x, \partial \boldsymbol{v}_{\boldsymbol{x}}\right), \boldsymbol{g}(\boldsymbol{x}, t) - \boldsymbol{v}(t; \boldsymbol{x})\}, \\
&\approx \min\{\partial \boldsymbol{v}^\delta_t(t; \boldsymbol{x}) + \boldsymbol{H}^\delta\left(t; x, \partial \boldsymbol{v}^\delta_{\boldsymbol{x}}\right), \boldsymbol{g}(\boldsymbol{x}, t) - \boldsymbol{v}^\delta(t; \boldsymbol{x})\}, \\
&:= \min\{\partial \boldsymbol{v}^\delta_t(t; \boldsymbol{x}) + \max_{\boldsymbol{u} \in \mathcal{U}} \min_{\boldsymbol{w} \in \mathcal{W}} \langle f(t; \boldsymbol{x}, \boldsymbol{u}, \boldsymbol{w}), \partial \boldsymbol{v}^\delta_{\boldsymbol{x}} \rangle, \\
&\qquad \boldsymbol{g}(t; \boldsymbol{x}) - \boldsymbol{v}^\delta(t; \boldsymbol{x})\}.
\end{aligned} \tag{17}$$

Thus, evaluating the spatial gradient of $\boldsymbol{v}^\delta$ in (17) is tantamount to drawing samples using the proximity expression above and using (6c) to evaluate $\boldsymbol{v}^\delta$. We can now write the variational HJ-Isaacs inequality (HJI-RCBRAT) as

$$\begin{aligned}
\min\Big\{ &\partial \boldsymbol{v}^\delta_t(t; \boldsymbol{x}) + \max_{\boldsymbol{u} \in \mathcal{U}} \min_{\boldsymbol{w} \in \mathcal{W}} \Big\langle f(t; \boldsymbol{x}, \boldsymbol{u}, \boldsymbol{w}), \\
&\frac{1}{t}(\boldsymbol{x} - \text{Prox}_{t\boldsymbol{g}}(\boldsymbol{x})) \Big\rangle, \qquad \qquad (18a) \\
&\boldsymbol{g}(\boldsymbol{x}, t) + \delta \log \mathbb{E}_{\boldsymbol{y} \sim \mathcal{N}(\boldsymbol{x}, \delta t)} \left[\exp(-\delta^{-1} \boldsymbol{g}(\boldsymbol{y})) \right] \Big\} \leq 0 \\
&\boldsymbol{v}(0; \boldsymbol{x}) = \boldsymbol{g}(0; \boldsymbol{x}). \qquad \qquad (18b)
\end{aligned}$$

D. Reducing Sampling Complexity

We now cast the solution of the variational HJI computational problem as a Monte Carlo algorithmics framework. Looking at equation (15), since we are working in a sampling-based values spatial derivative setting, we must avoid using too large samples to mitigate against time inefficiency while solving the problem.

While a weighted moving average of the gradient was mentioned in [11], no incorporation of prior data was imposed on the computational scheme. In this work, we incorporate a data-driven prior to every successive sample by first fitting Gaussian models to the numerator and denominator in (15). Preventing large samples being used to compute the proximal operator then amounts to conditioning the Gaussian model as ($\text{Prox}_{(t+1)\boldsymbol{g}} | \text{Prox}_{t\boldsymbol{g}}$) at each time step t. This is similar to linear regression with the advantage of allowing us to integrate prior information in the form of a Gaussian-inverse-Wishart distribution model.

For an *equivalent sample size* κ_0 and its belief, ν_0 on the prior variance, Σ_0 that is associated with the precision $\Lambda_0 = \Sigma_0^{-1}$ of a dataset D, suppose that the mean and variance of the normal-inverse-Wishart natural conjugate prior [27] are,

$$\Sigma \sim IW_{\nu_0}(\Lambda_0^{-1}), \quad \mu \mid \Sigma \sim \mathcal{N}(\mu_0, \Sigma/\kappa_0) \tag{19}$$

with associated empirical estimates $\hat{\Sigma}$ and $\hat{\mu}$. Then, the natural conjugate prior of the normal-inverse-Wishart is given as $p(\mu, \Sigma) \triangleq NIW(\mu_0, \kappa_0, \Lambda_0, \nu_0)$. We are concerned with its maximum-a-posteriori

$$\begin{aligned}
&p(\mu, \Sigma | D, \mu_0, \kappa_0, \Lambda_0, \nu_0) = NIW(\mu, \Sigma | \mu_n, \kappa_n, \Lambda_n, \nu_n), \\
&\kappa_n = \kappa_0 + n, \ \nu_n = \nu_0 + n
\end{aligned} \tag{20}$$

where

$$\mu_n = \frac{\kappa_0 \mu_0}{\kappa_0 + n} + \frac{n}{\kappa_0 + n} \hat{\mu}, \tag{21a}$$

$$\Lambda_n = \Lambda_0 + S + \frac{\kappa_0 n (\kappa_0 + n)(\hat{\mu} - \mu_0)(\hat{\mu} - \mu_0)^\top}{\kappa_0 + n}, \tag{21b}$$

$$S = \sum_{i=1}^{n} (\boldsymbol{x} - \bar{\boldsymbol{x}})(\boldsymbol{x} - \bar{\boldsymbol{x}})^\top. \tag{21c}$$

We recursively incorporate new samples by leveraging the MAP parameters given in (21) to new samples drawn from (15).

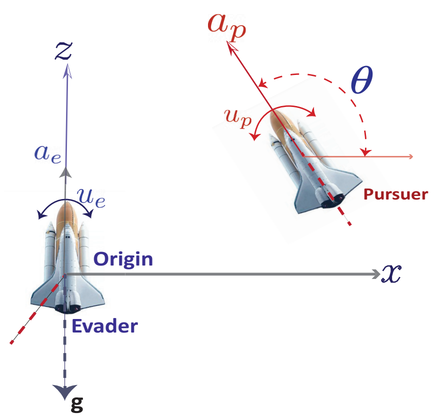

Fig. 1. Motion of two rockets on a Cartesian $\boldsymbol{xz}$-plane with a thrust inclination in relative coordinates given by $\theta := u_p - u_e$.

IV. Numerical Results

In this section, we provide a representative problem and amend it to the viscous HJI PDE (HJI-RCBRT) to test and verify our hypothesis.

A. The Rockets Launch Problem

We consider the rocket launch problem of Dreyfus [28] which was amended to a differential game between two identical rockets, $\boldsymbol{P}$ and $\boldsymbol{E}$, on an (x, z) cross-section of a Cartesian plane in [5], [6]. We want to compute the usable part [16] or RCBRT [4] of the *approximate* terminal surface's boundary for a predefined target set over a target tube. The usable part entails the regions of the state-space for which the min-max operation over either strategy of $\boldsymbol{P}$ and $\boldsymbol{E}$ is below zero and where the boundary of the usable part (BUP) constitute where (HJI-RCBRAT) is exactly zero.

For a two-player differential game, let $\boldsymbol{P}$ and $\boldsymbol{E}$ share identical dynamics in a general sense so that we can freely choose the coordinates of $\boldsymbol{P}$; however, $\boldsymbol{E}$'s origin is a distance Φ away from (x, z) at plane's origin (see Fig. 1) so that the $\boldsymbol{PE}$ vector's inclination measured counterclockwise from the $\boldsymbol{x}$ axis is θ.

Hence, we may consider the current problem as a terminal pursuit-evasion *differential* game, whereupon the game terminates when *capture* occurs. This eventual terminal represents the final *overapproximated* (non)-capture surface for the $\boldsymbol{P}-\boldsymbol{E}$ game (to be shortly introduced). We desire that every `max` and `min` operations during a *game of kind* are in the interior of the plane where motion exists; this way, no sudden changes from extremes are too aggravating in cost.

Let the states of $\boldsymbol{P}$ and $\boldsymbol{E}$ be $(\boldsymbol{x}_p, \boldsymbol{x}_e)$, respectively, driven by thrusts (u_p, u_e) respectively (see Figure 1). Fix the rockets' range so that what is left of the motion of either $\boldsymbol{P}$ or $\boldsymbol{E}$'s is restricted to orientation on the (x, z) plane as illustrated in Fig. 1. where a and g are respectively the acceleration and gravitational accelerations (in feet per square second) [4]. As long as $\boldsymbol{E}$ remains within this target region or *backward reachable tube* (or BRT), $\boldsymbol{P}$ cannot cause damage or exercise an action of deleterious consequence on, say, the territory being guarded by $\boldsymbol{E}$. Setting up $\boldsymbol{E}$ to maximize a payoff quantity with the largest possible margin or at least frustrate the efforts of $\boldsymbol{P}$ with minimal collateral damage while the pursuer minimizes this quantity constitutes a terminal value *optimal* differential game: there is no optimal pursuit without an optimal evasion.

[4]We set $a = 1ft/sec^2$ and $g = 32ft/sec^2$ in our simulation.

The motion of $\boldsymbol{P}$ relative to $\boldsymbol{E}$'s along the $(\boldsymbol{x}, \boldsymbol{z})$ plane includes the relative orientation, the control input, shown in Fig. 1 as $\theta = u_p - u_e$. Following the conventions in Fig. 1, the game's relative equations of motion in *reduced space* [16, §2.2] was derived as (see [5, §VI.A]) is $\boldsymbol{x} = (x, z, \theta)$ where $\theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right)$ and $(x, z) \in \mathbb{R}^2$ are

$$\dot{\boldsymbol{x}} = \begin{cases} \dot{x} & = a_p \cos\theta + u_e x, \\ \dot{z} & = a_p \sin\theta + a_e + u_e x - g, \\ \dot{\theta} & = u_p - u_e. \end{cases} \tag{22}$$

The capture radius of the origin-centered circle Φ (we set $\boldsymbol{g} = 1.5$ ft) is $\|\boldsymbol{PE}\|_2$ so that $\boldsymbol{g}^2 = x^2 + z^2$. All capture points are specified by the variational HJ PDE [4]:

$$\frac{\partial \boldsymbol{g}}{\partial t}(\boldsymbol{x}, t) + \min\left[0, \boldsymbol{H}\left(\boldsymbol{x}, \frac{\partial \boldsymbol{g}(\boldsymbol{x}, t)}{\partial \boldsymbol{x}}\right)\right] \leq 0, \tag{23}$$

with Hamiltonian given by

$$\begin{aligned} \boldsymbol{H}(\boldsymbol{x}, p) = & -\max_{u_e \in [\underline{u}_e, \bar{u}_e]} \min_{u_p \in [\underline{u}_p, \bar{u}_p].} \begin{bmatrix} p_1 & p_2 & p_3 \end{bmatrix} \\ & \begin{bmatrix} a_p \cos\theta + u_e x \\ a_p \sin\theta + a_e + u_p x - g \\ u_p - u_e \end{bmatrix}. \end{aligned} \tag{24}$$

Here, p are the co-states, and $[\underline{u}_e, \bar{u}_e]$ denotes extremals that the evader must choose as input in response to the extremal controls that the pursuer plays i.e. $[\underline{u}_p, \bar{u}_p]$. Rather than resort to analytical *geometric reasoning*, we may analyze possibilities of behavior by either agent via a principled numerical simulation. This is the essence of this work. From (24), set $\underline{u}_e = \underline{u}_p = \underline{u} \triangleq -1$ and $\bar{u}_p = \bar{u}_e = \bar{u} \triangleq +1$ so that $\boldsymbol{H}(\boldsymbol{x}, p)$ is

$$\begin{aligned} & -\max_{u_e \in [\underline{u}_e, \bar{u}_e]} \min_{u_p \in [\underline{u}_p, \bar{u}_p]} \begin{bmatrix} p_1(a_p \cos\theta + u_e x) + \\ p_2(a_p \sin\theta + a_e + \\ u_p x - g) + p_3(u_p - u_e) \end{bmatrix} \\ & \triangleq -a p_1 \cos\theta - p_2(g - a - a\sin\theta) - \bar{u}|p_1 x + p_3| \\ & \quad + \underline{u}|p_2 x + p_3|, \end{aligned} \tag{25}$$

where the last line follows from setting $a_e = a_p \triangleq a$.

B. Numerical Integration Considerations

We take the infinitesimal physics problem above and make it algorithmic by discretizing the spatial and temporal domains i.e.

$$(x, z) \in (-100, +100],\ \theta \in (-\pi/2, \pi/2],\ t \in (0, 1], \tag{26}$$

with the resolutions $t_k - t_{k-1} = 1/1000$ along the temporal dimension and $\boldsymbol{x}_k - \boldsymbol{x}_{k-1} = 1/100$ along the spatial dimensions, where $k \in \mathbb{N}$.

Care is taken not to combine the state components into one full spatial domain early on in the algorithm. We first sampled from each spatial component of (26), compute the values within the expectations in (15), and then compute the proximal for the full state space using the expectation-maximization algorithm [29]. The expectation-maximization algorithm proceeded as follows:

- We first computed the logarithm of the observations

We do not employ a mesh or grid since the indices from which we sample are uniform across x, z and θ. The innards of the expression in (15) were first computed based on samples from the state space (26).

V. Conclude

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